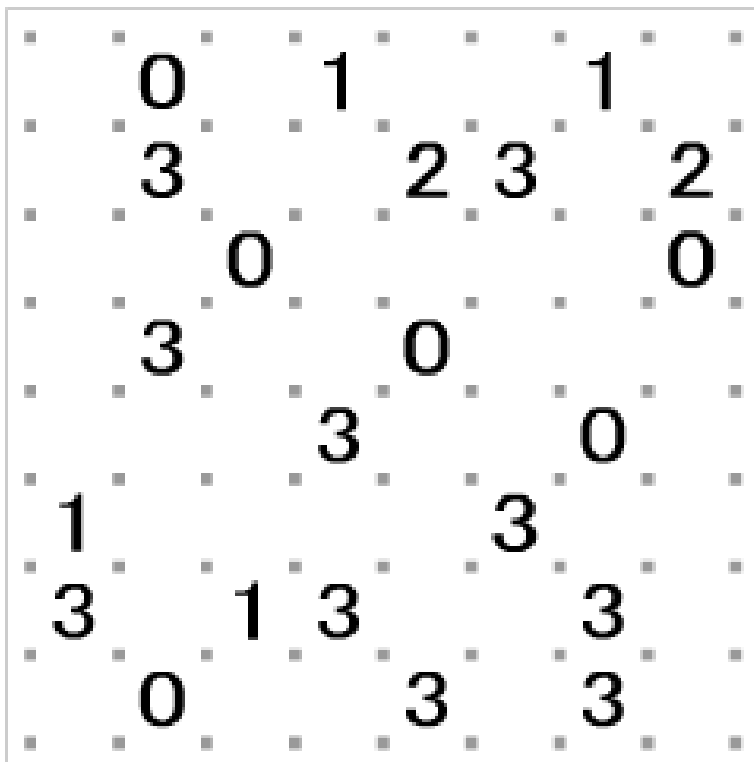
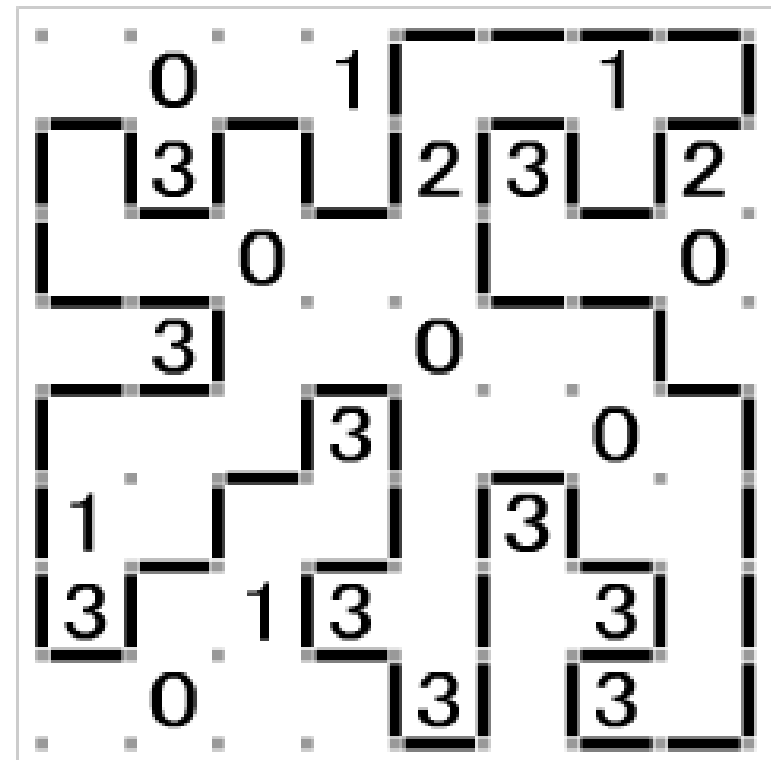


Puzzle: Slitherlink



Input problem



A solution

Slitherlink Encoding

Assume that x_1, x_2, x_3, x_4 are 4 vertices around position i, j . If value at i, j is

- 0: $\neg x_1 \wedge \neg x_2 \wedge \neg x_3 \wedge \neg x_4$
- 1: $(x_1 \vee x_2 \vee x_3 \vee x_4) \wedge$
 $(x_1 \rightarrow \neg x_2 \wedge \neg x_3 \wedge \neg x_4) \wedge$
 $(x_2 \rightarrow \neg x_1 \wedge \neg x_3 \wedge \neg x_4) \wedge$
 $(x_3 \rightarrow \neg x_2 \wedge \neg x_1 \wedge \neg x_4) \wedge$
 $(x_4 \rightarrow \neg x_2 \wedge \neg x_3 \wedge \neg x_1)$

Slitherlink Encoding (cont)

Assume that x_1, x_2, x_3, x_4 are 4 vertices around position i, j . If value at i, j is

- 2: $(x_1 \vee x_2 \vee x_3) \wedge (x_1 \vee x_2 \vee x_4) \wedge (x_1 \vee x_3 \vee x_4) \wedge (x_2 \vee x_3 \vee x_4) \wedge (\neg x_1 \vee \neg x_2 \vee \neg x_3) \wedge (\neg x_1 \vee \neg x_2 \vee \neg x_4) \wedge (\neg x_1 \vee \neg x_3 \vee \neg x_4) \wedge (\neg x_2 \vee \neg x_3 \vee \neg x_4)$

Slitherlink Encoding (cont)

Assume that x_1, x_2, x_3, x_4 are 4 vertices around position i,j . If value at i,j is

- 3: $(\neg x_1 \vee \neg x_2 \vee \neg x_3 \vee \neg x_4) \wedge$
 $(\neg x_1 \rightarrow x_2 \wedge x_3 \wedge x_4) \wedge$
 $(\neg x_2 \rightarrow x_1 \wedge x_3 \wedge x_4) \wedge$
 $(\neg x_3 \rightarrow x_2 \wedge x_1 \wedge x_4) \wedge$
 $(\neg x_4 \rightarrow x_2 \wedge x_3 \wedge x_1)$
- 4: $(x_1 \wedge x_2 \wedge x_3 \wedge x_4)$